

Data Analysis System

I. Financial Data Analysis System

1. Data Ingestion Layer

- 1.1 User Dataset Upload
- 2.2 Data Ingestion Agent
- 3.3 Schema Validation
- 4.4 Source Understanding
- 5.5 Analysis Router Agent

2. Numerical Data Pipeline

- 1.1 Datatype Validation
 - Numeric Stored as String Detection
 - Currency Formatting Validation
 - Decimal Precision Validation
- 2.2 Missing Value Analysis
 - Null Detection
 - MCAR/MAR/MNAR Classification
 - Median Imputation
 - Mean Imputation
- 3.3 Duplicate Detection
- 4.4 Outlier Detection
 - IQR Method
 - Z-score Method
 - Isolation Forest
 - Anomaly Flagging
 - Fraud Signal Detection
- 5.5 Distribution Analysis
 - Normal Distribution
 - Skewed Distribution
 - Bimodal Distribution
 - Heavy-tail Distribution
 - KDE Analysis
 - Histogram Analysis
- 6.6 Statistical Moments
 - Mean

- Median
- Mode
- Variance
- Standard Deviation
- Skewness
- Kurtosis
- Percentile Analysis

7.7 Correlation Analysis

- Pearson Correlation
- Spearman Correlation
- Covariance
- Multicollinearity Detection
- VIF Analysis

8.8 Scaling & Normalization

- StandardScaler
- RobustScaler
- MinMaxScaler

9.9 Feature Engineering

- Ratio Creation
- Log Transformation
- Rolling Statistics
- Interaction Features

10.10 Financial Analysis

- Revenue Trends
- Profit Analysis
- Risk Metrics
- VaR
- CVaR
- Sharpe Ratio
- Anomaly Transaction Analysis

11.11 Visualization

- Histogram
- Boxplot
- Scatterplot
- Violin Plot
- Heatmap
- Line Chart

12.12 Numerical Insights

3. Categorical Data Pipeline

1.1 Category Validation

- Inconsistent Category Detection

- Typo Detection
- Category Standardization
- 2.2 Missing Value Handling
 - Mode Imputation
 - Unknown Category Handling
- 3.3 Cardinality Analysis
 - Low Cardinality Detection
 - High Cardinality Detection
- 4.4 Frequency Analysis
 - Value Counts
 - Category Dominance
 - Rare Category Detection
- 5.5 Imbalance Analysis
 - Class Imbalance Detection
 - Distribution Percentage Analysis
- 6.6 Relationship Analysis
 - Grouped Aggregation
 - Categorical Dependency
 - Chi-square Analysis
- 7.7 Encoding Techniques
 - One-hot Encoding
 - Label Encoding
 - Target Encoding
 - Hash Encoding
- 8.8 Segmentation Analysis
 - Customer Segmentation
 - Branch Segmentation
 - Product Segmentation
- 9.9 Visualization
 - Countplot
 - Bar Chart
 - Pie Chart
 - Stacked Bar Chart
- 10.10 Categorical Insights
- 4. Datetime Data Pipeline**
 - 1.1 Datetime Validation
 - 2.2 Missing Date Analysis
 - 3.3 Trend Analysis
 - 4.4 Seasonality Analysis
 - 5.5 Rolling Averages
 - 6.6 Time Decomposition

- 7.7 Forecast Preparation
- 8.8 Time-based Feature Engineering
- 9.9 Month-over-month Analysis
- 10.10 Year-over-year Analysis
- 11.11 Time-series Visualization
- 12.12 Temporal Insights

5. Text Data Pipeline

- 1.1 Text Cleaning
- 2.2 Tokenization
- 3.3 Stopword Removal
- 4.4 Lemmatization
- 5.5 Sentiment Analysis
- 6.6 Keyword Extraction
- 7.7 Topic Modeling
- 8.8 TF-IDF Analysis
- 9.9 Embedding Generation
- 10.10 Complaint Analysis
- 11.11 NLP Insights

6. Advanced Enterprise Features

- 1.1 Data Quality Scoring
- 2.2 Data Drift Detection
- 3.3 Explainable AI
- 4.4 Root Cause Analysis
- 5.5 Audit Logging
- 6.6 Regulatory Compliance
- 7.7 Fraud Detection
- 8.8 Real-time Monitoring
- 9.9 Alert Generation
- 10.10 Human-in-the-loop Validation
- 11.11 Role-based Access Control
- 12.12 Dashboard Generation
- 13.13 Automated Reporting
- 14.14 Recommendation Engine

7. Output Layer

- 1.1 Insight Generation Agent
- 2.2 Visualization Agent
- 3.3 Reporting Agent
- 4.4 Dashboard Layer

8. User Query Response Agent